Full-comb cleanup

the r457 race table was a truncated-comb echo

Stefan Hamann

August 30, 2026

Abstract

r458 showed that the r457 race values at k_z197 (0.996) and k_z230 (2.165) ran on the truncated campaign table (TABLE_CAP $= 4 \cdot 10^5$). Recomputing the r457 budget $q^\dagger_{\text{full}} = q_* + \Delta q_{\text{arith}}$, $q_* = q^\dagger_{\text{ARCH}}(J_P)$, race $= \Delta q_{\text{arith}}/(1 - q_*)$ on the complete a^2 comb breaks the monotone climb: after k_z136 the race falls back into 0.72–0.77. On the Lean selected family the race stays in 0.61–0.76 through $k = 12$ (exact dps = 50). k_z137 still dies with a complete comb, but the Lean-admissible window at the same $a = 641$ lives — the residual death is outside the mincut object. No RH claim.

Status. No new SATZ. Ausgang `RACE_TREND_BROKEN / LEANFAM_EXACT_ALIVE(12) / KZ137_OUTSIDE_MINCU` The r457 identity $q^\dagger_{\text{full}} < 1 \iff \Delta q_{\text{arith}} < 1 - q_*$ stands. The r457 verdict `RACE_EATS_K10` does not. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.

1 Input completeness

Lemma 1 (comb inventory)**.** *An atom count k_a is accepted only if $k_a = \#\{p^\ell \le a^2\}$ and the sieve reaches a^2. This is the `ExactPrimeSource`(a) cutoff (prime powers, not $\pi(a^2)$). Reflection tents are already in the three-lag identity. Powers $p^\ell > a^2$ are outside the object; their absence is not an artefact inside the object.*

2 Frame-A race, completed

Same ARCH fold as r456/r457 (`measures_from_c`, c^A only versus $c^A + c^P$).

kz	a	q_{full}	q_*	Δq	race	r457 race	live
17	32	0.89136	0.78566	0.10570	0.493	0.493	Y
69	256	0.95251	0.89689	0.05562	0.539	0.539	Y
116	512	0.96221	0.86848	0.09374	0.713	0.713	Y
136	631	0.97209	0.87092	0.10117	0.784	0.784	Y
138	643	0.97342	0.88949	0.08393	0.759	—	Y
197	1024	0.96910	0.88786	0.08124	0.724	0.996	Y
230	1259	0.97090	0.87326	0.09764	0.770	2.165	Y

kz17/69/116/136 were already complete under the campaign table. kz197/230 were not.

Lemma 2 (trend broken)**.** *Every completed race is strictly below* 1*. Monotonicity fails after kz*136*:* $0.784 \to 0.760 \to 0.724 \to 0.770$*. AIC on the seven points prefers M_2. Dropping* 197*,* 230*, or* 138 *still prefers M_2. The old four-point ladder* $(17, 69, 116, 136)$ *prefers M_1 — that was the truncated chart.* `RACE_TREND_BROKEN`*.*

3 Lean family

The load-bearing ladder is the named selected sequence, not frame-A. $J_P = 3$ on $k = 5..8$ forces a pack with $n_\star \geq 2$ (the r451 gate $n_\star \geq 4$ is a chart convention, not a Lean constraint).

k	a	m	N	J_P	race	exact	complete
5	32	19	10	3	0.675	float	Y
6	64	23	12	3	0.695	float	Y
7	128	27	14	3	0.629	float	Y
8	256	31	16	3	0.607	float	Y
9	512	71	36	7	0.635	float	Y
10	1024	79	40	7	0.758	dps50 live	Y
11	2048	87	44	7	0.716	dps50 live	Y
12	4096	95	48	7	0.694	dps50 live	Y
14	16384	111	56	7	—	—	N ($a^2 = 2.68 \cdot 10^8$)

Lemma 3 (Lean race stays off 1)**.** *On every complete member $k = 5..12$ the race lies in $[0.607, 0.758]$. The sequence is not monotone and does not run toward 1. Exact* dps $= 50$ *confirms $n_{\text{flip}} = 0$ at $k = 10, 11, 12$ ($k_a = 82267, 296347, 1078555$). $k = 14$ is a sieve-budget limit, not a death.* `LEANFAM_EXACT_ALIVE(12)`.

This is the central number of the round: on the mincut-relevant family the arithmetic increment does not eat the Archimedean margin.

4 kz137

Lemma 4 (outside the mincut)**.** *kz137 ($a = 641$, $N = 8300$): inventory $k_a = 34856 = \#\{p^\ell \leq 410881\}$. The frame-A pack dies (mass-chain first flip 7308, r458). The Lean-admissible window at the same anchor $a = 641$ with selectedMesh $k = 9$ ($m = 71$, $N = 36$) and $k = 10$ ($m = 79$, $N = 40$) is living (race 0.453 / 0.534). Frame-A couples the mesh to the local log-gap and therefore consumes an extreme depth the Lean mincut never asks for.* `KZ137_OUTSIDE_MINCUT`.

5 Kill table

Kill	Status
Comb inventory $k_a = \#\{p^\ell \leq a^2\}$	sealed on every scored window
ARCH fold identity (r456)	used throughout
Exact dps $= 50$	Lean $k = 10, 11, 12$ $n_{\text{flip}} = 0$
Determinism	two identical kz17 races
Fit transparency	full 7: M_2; only-old: M_1
$k = 14$ exact / complete	`BUDGET_LIMIT` (honest)

Claim boundary. Finite-window cleanup census. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.